Date.

C

C12P1

T1 C

T2 0，变小

T3 30°，150°

T4 B

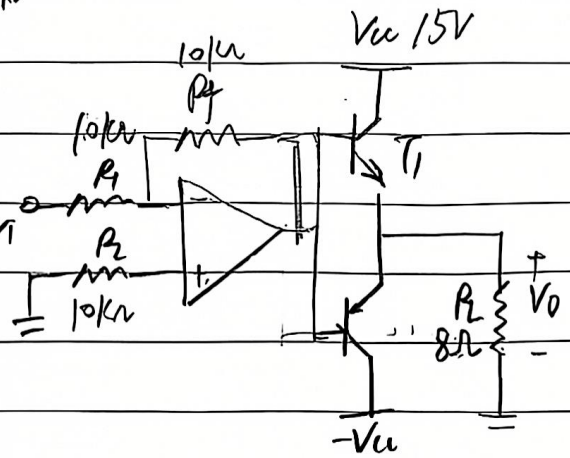
T5 B

T6 B

T7 B

C12P2

T1



T2 C

T3 $V_o = V_{cc} - |V_{CES}| = 14V$

$P_{Dmax} = \frac{V_{om}^2}{2R_L} = 24.5W$

$P_C = 2V_{cc} \cdot \frac{1}{2\pi}\int_0^{\pi} \frac{V_{om}\sin\theta}{R_L} d\theta = \frac{2}{\pi}\frac{V_{cc}}{R_L}V_o$

$\eta = \frac{P_L}{P_S} = \frac{\pi}{4}\frac{V_o}{V_{cc}} = 0.687$

T4 $P_D = P_S - P_L = \frac{2V_{cc}}{\pi R_L}V_o - \frac{V_o^2}{2R_L}$

$\frac{dP_D}{dV_o} = \frac{2V_{cc}}{\pi R_L} - \frac{V_o}{R_L} = 0 \Rightarrow V_o = \frac{2}{\pi}V_{cc}$

代入，得 $P_D = \frac{2}{\pi^2}\frac{V_{cc}^2}{R_L}$

故 $P_{Tmax} = \frac{1}{2}P_D = \frac{1}{\pi^2}\frac{V_{cc}^2}{R_L} = 6.485W$

$\bar{V} = 9.90V$

(1) 电压并联负反馈

(2) $P_{Lmax} = \frac{V_{om}^2}{2R_L} = 24.625W$

(3) $P_S = 2V_{cc}\frac{1}{2\pi}\int_0^{\pi}\frac{\hat{V}_o\sin\theta}{R_L}d\theta = \frac{2}{\pi}\frac{V_{cc}}{R_L}\hat{V}_o$ $P_D = P_S - P_L = \frac{2}{\pi}\frac{V_{cc}}{R_L}V_o - \frac{V_o^2}{2R_L}$

$\frac{dP_D}{d\hat{V}_o} = \frac{2}{\pi}\frac{V_{cc}}{R_L} - \frac{\hat{V}_o}{R_L} = 0 \Rightarrow \hat{V}_o = \frac{2}{\pi}V_{cc}$ $P_{Cm} = \frac{1}{2\pi^2}\frac{V_{cc}^2}{R_L} = 2.85W$

$V_{BR(CEO)} = 2V_{CC} = 30V$

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$P_S = \frac{2V_{cc}}{2R_L} V_0$

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T5 (1) $P_{om} = \frac{(V_{cc} - |V_{CES}|)^2}{R_L} = 8W$

$\eta = \frac{1}{4}\frac{V_0}{V_{cc}} = 0.698$

(2) $I_{max} = \frac{V_{cc} - |V_{CES}|}{R_L} = 0.5A$

$V_{max} = 2V_{cc} - I_{max} = 34V$

$P_{S max} = \frac{2V_{cc}^2}{2R_L} = 6.45W \Rightarrow P_T = \frac{1}{5} P_{S max} = 3.23W$

T6 $P_{om} = \frac{V_{cc}^2}{2R_L} = 14.063W$

$P_S = 2V_{cc} \frac{1}{2\pi} \int_0^{\pi} \frac{V_0 \sin\theta}{R_L} d\theta = \frac{2V_{cc}}{2R_L} V_0 = \frac{2V_{cc}^2}{2R_L} = 17.905W$

$\eta = \frac{P_L}{P_S} = 0.785$

T7 (1) $|V_{CES}| = 5V$

$P = \frac{(V_{cc} - V_{CES})^2}{2R_L} = 20W \Rightarrow V_{cc} = 22.9V$

(2) $I = \frac{V_{cc} - V_{CES}}{R_L} = 2.24A$

(3) $P_S = 2V_{cc} \frac{1}{2\pi} \int_0^{\pi} \frac{V_0 \sin\theta}{R_L} d\theta = \frac{2V_{cc}}{2R_L} V_0 = 32.62W$

(4) $\eta = \frac{P_L}{P_S} = 0.613$

(4) $P_D = \frac{P_S - P}{2} = 6.31W$

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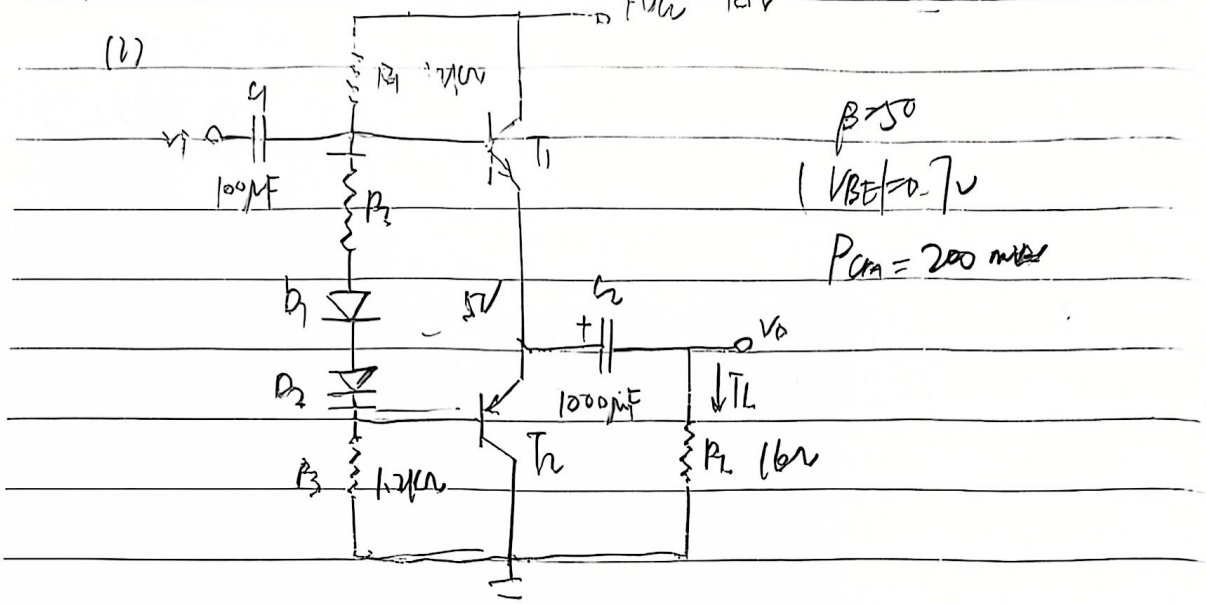
8 Ω, $\frac{P_L}{P_D} \geq \frac{\eta}{\eta_0} \Rightarrow P_D = \frac{1}{\eta} = 1.1W$

此时 $P_{om} = \frac{V_{cc}^2}{2R_L} = 4.84W$

$\eta = \frac{\pi}{4}\frac{V_{cm}}{V_{cc}} = 73.7\%$ 此时 $V_{cc} = \frac{2V_{cc}}{\pi+1} V_c = 11.1W$

$P_D = \frac{1}{\eta}(P_{om} - P_L) = 2.35W$

T9 (1) 调大 R_2



$\beta = 50$

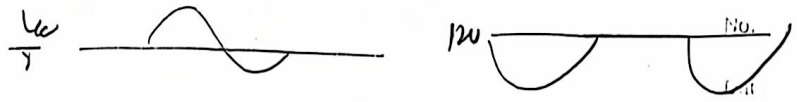
$|V_{BE}| = 0.7V$

$P_{CM} = 200mW$

(1) $I_{B1} = I_{B2} = \frac{V_{cc} - 2V_{BE}}{R_1 + R_3} = 2.58mA$

$I_c = \beta I_b = 179.17mA$

$P = I_c \frac{V_{cc}}{2} = 895.9mW > P_{om}$，会烧毁



IV (1) $P_O = \frac{1}{2\pi}\int_0^{2\pi} i_C^2 \, d\omega t \, R_L = 3.54W$ $\left(\frac{I_{cm}^2}{2} R_L\right)$

(2) $P_E = \frac{1}{2\pi} V_{CC} \int_0^{2\pi} |i_C| \, d\omega t = 5.01W$

(3) $P_T = P_E - P_O = 1.47W$

III (1) $V_{im} = \frac{V_{CC} - V_{CES}}{|2|A_V|} = 78mV$

II

(2) $V_O = |A_V| V_i = 1V$

~~$V_O = V_{CC} - \frac{1}{2\pi}\int_0^{2\pi} V_{CES} \sin\theta \, d\theta = 8.13V$~~ $V_O = 11.3V$

I (1) $I_{R2} = \frac{V_{BE} - V_{in}}{R_L} = 12.5mA$ $I_{Q1} = 20mA$

$I_{R2} = \frac{V_{BE} - V_{in}}{R_L} = 2.5mA$ $I_{Q1} = 7.5mA$

(2) $I_{Q1} \equiv 0.3mA$

$V_{out} = -0.99 I_1 R_L = -25.4mV$

$V_{in} = V_T \ln \frac{I_Q}{I_S} + V_{out} = 339.9mV$